

Todo morfismo de anillos de un cuerpo a un anillo no nulo es inyectivo

Lema 1. *Sea K un cuerpo, A un anillo (no nulo) y $\phi: K \rightarrow A$ un morfismo de anillos. Entonces ϕ es inyectivo.*

Demostración: Sean $a, b \in K$ tales que $\phi(a) = \phi(b)$, entonces

$$\phi(a) - \phi(b) = 0 \implies \phi(a - b) = 0 \implies a - b = 0 \implies a = b.$$

Por tanto, ϕ es inyectivo. ■

Referenciado en

- Extensión de cuerpos

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